

# Lecture 25: Linear Regression

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- 1 Linear regression as estimation::: {.callout-note icon=false}### Definition: Simple linear regressionModel  $Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ ,  $\varepsilon_i$  iid mean-zero. The least-squares estimator minimizes  $\sum_i (y_i - \beta_0 - \beta_1 x_i)^2$ :

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

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- 1.1 Reading & HomeworkReading: §9, supplementary notes (Bertsekas & Tsitsiklis, Introduction to Probability). . . . . 2

—title: “Lecture 25: Linear Regression”author: “Jamie Haddock”format: revealjs: output-file: Lecture25\_slides slide-number: true preview-links: auto logo: figs/hmc.png css: input/slides.css incremental: true smaller: true code-fold: true embed-resources: true html: code-fold: true embed-resources: true pdf: documentclass: article toc: true number-sections: true geometry: - top=1in - left=1in - bottom=1in - right=1informat-links: falsejupyter: python3filters: - input/remove-pause.luaexecute: echo: true eval: true—

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```
n = 60beta0_true, beta1_true, sigma = 2.0, 1.5, 1.0x = rng.uniform(0, 10, size=n)y = beta0_true + beta1_true * x + sigma * rng.standard_normal(size=n)from scipy.optimize import minimizedef neg_log_lik(params):    b0, b1, log_sigma = params    sigma = np.exp(log_sigma)
```

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